

Higher Diploma in Software Engineering – HDSE23.1F/CO

Machine Learning for Artificial Intelligence

Time allowed: 9:00am – 12:00am (3 hours)

08th of April 2024

INSTRUCTIONS TO CANDIDATES

- This paper contains 5 questions.
- Answer **ALL** questions.
- The total marks obtainable for this examination is **100**.
- Marks for each question are indicated.
- This examination accounts for 50% of the course assessments.
- This is an **Open book** examination.
- All calculators are **not** allowed.

Question 1

(Marks 25)

- Explain / describe what Machine Learning is? (Marks 6)
- List important reasons for why Machine Learning is used. (Marks 6)
- Explain the main three categories of Machine Learning applications. (Marks 6)
- Identify three practical applications of Machine Learning. (Marks 7)

Question 2

(Marks 25)

- What is supervised machine learning? (Marks 7)
- Describe the operational mechanism of one regression technique commonly used in Machine Learning with proper examples. (Marks 9)
- Describe the operational mechanism of one classification technique commonly used in Machine Learning with proper examples. (Marks 9)

Question 3**(Marks 25)**

- a. Describe and illustrate the operational process of Support Vector Machine (SVM) with a suitable example. (Marks 6)
- b. Explain the differences between K-mean and K-nearest neighbors (KNN) algorithms. (Marks 6)
- c. Describe the operational mechanism of the K-nearest neighbors (KNN) algorithm in Machine Learning. (Marks 6)
- d. Describe the operational mechanism of the Random Forest algorithm in Machine Learning. (Marks 7)

Question 4**(Marks 25)**

- a. Describe the rationale behind the necessity of preprocessing or cleaning the data in the context of Machine Learning. (Marks 6)
- b. Describe all the steps involved in training a Machine Learning model. (Marks 6)
- c. Describe the process for selecting the optimal model from a set of trained Machine Learning models. (Marks 6)
- d. Differentiate between standardization and normalization in the context of data preprocessing. (Marks 7)